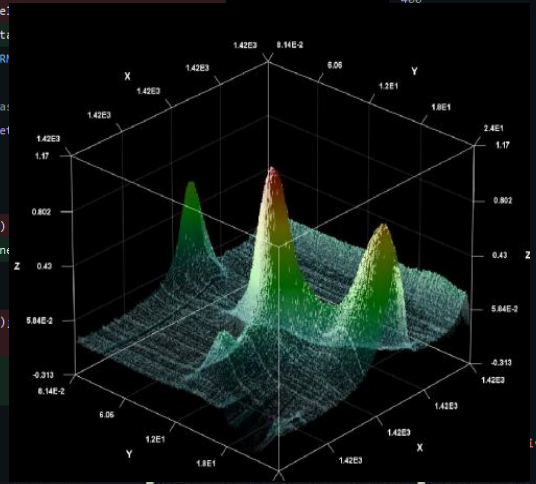
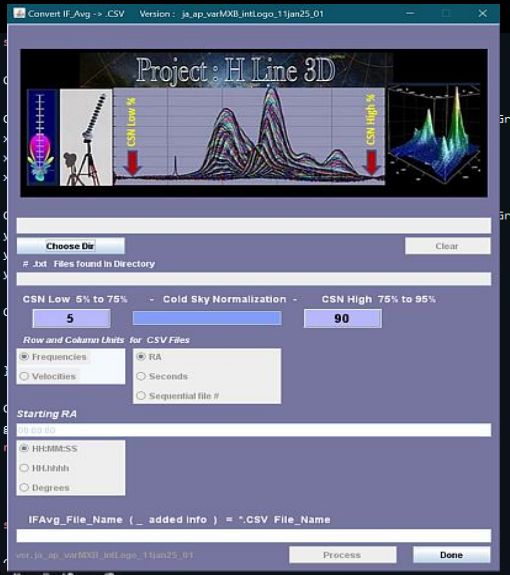
File Conversion & Spectral Processing Software

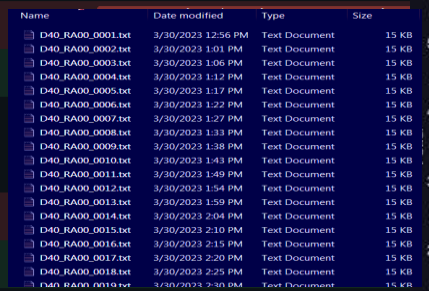
```
raDegreesRadioButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        dataCruncher.rightAscensionCalculator.setInitialRa(raStartTextFie
        dataCruncher.rightAscensionCalculator.setInitialRaFromString(raSt
        dataCruncher.rightAscensionCalculator.setRaFormat(RaFormat.RA_FOR
        //raStartTextField.setFormatterFactory(new
DefaultFormatterFactory(dataCruncher.rightAscensionCalculator.getRaFormat().getMa
        raStartTextField.setText(dataCruncher.rightAscensionCalculator.get
    }
});

outfileTextField.getDocument().addDocumentListener(new DocumentListener()
sessionNameTextField.getDocument().addDocumentListene
    @Override
    public void removeUpdate(DocumentEvent e) {
        System.out.println("Outfile text: " + outfileTextField.getText());
        dataCruncher.setOutfileName(outfileTextField.getText());
        sessionName = sessionNameTextField.getText();
        System.out.println("Session name: " + sessionName);
    } // removeUpdate()

    @Override
    public void insertUpdate(DocumentEvent e) {
```

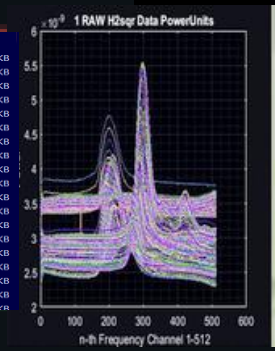


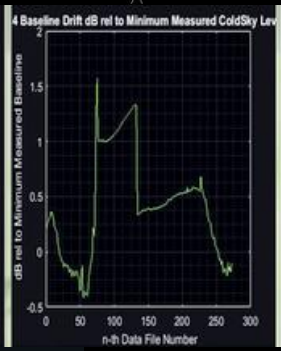


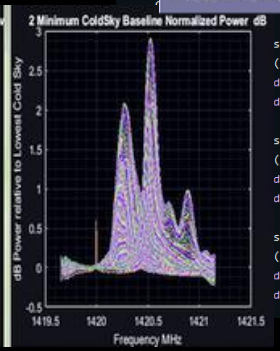


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D40_RA00_0002.txt	3/30/2023 1:01 PM	Text Document	15 KB
D40_RA00_0003.txt	3/30/2023 1:06 PM	Text Document	15 KB
D40_RA00_0004.txt	3/30/2023 1:12 PM	Text Document	15 KB
D40_RA00_0005.txt	3/30/2023 1:17 PM	Text Document	15 KB
D40_RA00_0006.txt	3/30/2023 1:22 PM	Text Document	15 KB
D40_RA00_0007.txt	3/30/2023 1:27 PM	Text Document	15 KB
D40_RA00_0008.txt	3/30/2023 1:33 PM	Text Document	15 KB
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D40_RA00_0012.txt	3/30/2023 1:54 PM	Text Document	15 KB
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D40_RA00_0015.txt	3/30/2023 2:10 PM	Text Document	15 KB
D40_RA00_0016.txt	3/30/2023 2:15 PM	Text Document	15 KB
D40_RA00_0017.txt	3/30/2023 2:20 PM	Text Document	15 KB
D40_RA00_0018.txt	3/30/2023 2:25 PM	Text Document	15 KB

```
    } // changedUpdate()
});
```








```
sColorGradient xGradient = new ColorGradient.AxisColorGradient();
(ColorGradient.Axis.X);
daryColors(new Color[] {clearBlack, Color.RED});
dMode(ColorGradient.BlendMode.ADDITION);

sColorGradient yGradient = new ColorGradient.AxisColorGradient();
(ColorGradient.Axis.Y);
daryColors(new Color[] {clearBlack, Color.GREEN});
dMode(ColorGradient.BlendMode.ADDITION);

sColorGradient zGradient = new ColorGradient.AxisColorGradient();
(ColorGradient.Axis.Z);
daryColors(new Color[] {clearBlack, Color.BLUE});
dMode(ColorGradient.BlendMode.ADDITION);
```

Software

Processing and Graphics Software Installation

JRE : Java Runtime Environment (Windows)

Java version(s): 8.0 Platform(s):

Windows 11, Windows 10, Windows 8, Windows 7, Windows Vista, Windows XP,

https://www.java.com/en/download/help/windows_offline_download.html

install JRE

Pettit / Adcock : JAVA_HL3D .jar :

Conversion of a data set of IF_Ave *.txt files to Rinearn compatible *.csv files

https://github.com/AP-HLine-3D/HLine3D

download latest Revision of JAVA_HL3D_.jar and drag to Desktop

Rinearn 2D & 3D Graphics Software

https://www.rinearn.com/en-us/graph2d/
https://www.rinearn.com/en-us/graph3d/

download and install 2D and 3D

Software

Processing and Graphics Software Installation

Pettit / Adcock : JAVA_HL3D .jar :

Conversion of a data set of IF_Ave *.txt files to Rinearn compatible *.csv files

https://github.com/AP-HLine-3D/HLine3D

download latest Revision of JAVA_HL3D_.jar and drag to Desktop

NOTE : Certain Java interpreters configured for languages other than English,
Spanish for example, use Commas vs Periods to define the decimal number delineator.
Commas are not compatible with Rinearn Graphics software

Solution :

Launch the interpreter with a command that forces it to load the English format:

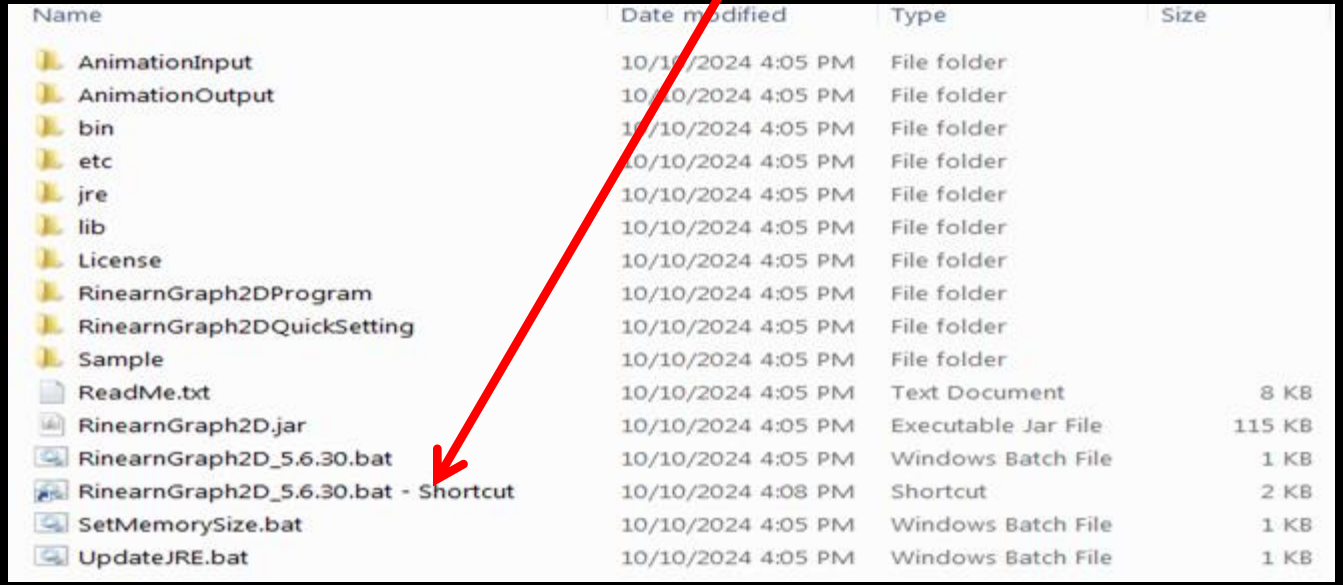
java "-Duser.language=en" "-Duser.country=US" -jar .JAVA_HL3D_33%Me-Mi_Vmxb_Vsplt_14May26.jar

(or the current JAVA_HL3D . jar version)

Graphics Software

Rinearn 2D Graphics Software

From installation subdirectory , create Shortcut to *RinearnGraph2D_5.6.30.bat*
and Drag Shortcut to Desktop

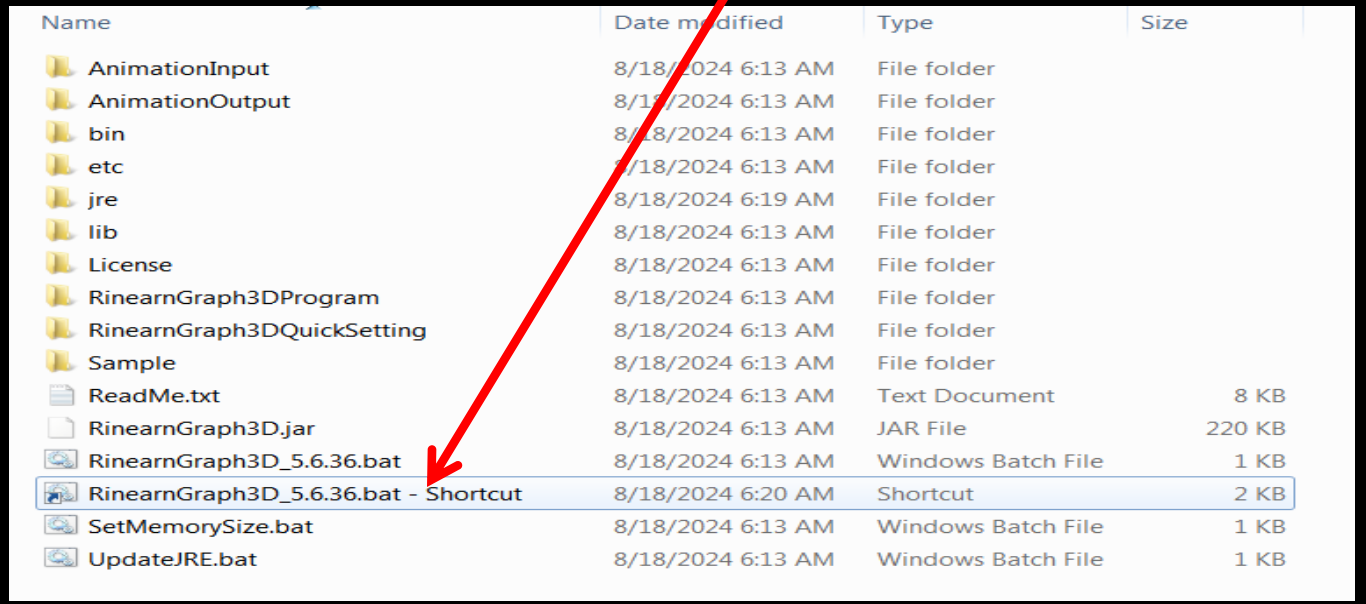


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AnimationOutput	10/10/2024 4:05 PM	File folder	
bin	10/10/2024 4:05 PM	File folder	
etc	10/10/2024 4:05 PM	File folder	
jre	10/10/2024 4:05 PM	File folder	
lib	10/10/2024 4:05 PM	File folder	
License	10/10/2024 4:05 PM	File folder	
RinearnGraph2DProgram	10/10/2024 4:05 PM	File folder	
RinearnGraph2DQuickSetting	10/10/2024 4:05 PM	File folder	
Sample	10/10/2024 4:05 PM	File folder	
ReadMe.txt	10/10/2024 4:05 PM	Text Document	8 KB
RinearnGraph2D.jar	10/10/2024 4:05 PM	Executable Jar File	115 KB
RinearnGraph2D_5.6.30.bat	10/10/2024 4:05 PM	Windows Batch File	1 KB
RinearnGraph2D_5.6.30.bat - Shortcut	10/10/2024 4:08 PM	Shortcut	2 KB
SetMemorySize.bat	10/10/2024 4:05 PM	Windows Batch File	1 KB
UpdateJRE.bat	10/10/2024 4:05 PM	Windows Batch File	1 KB

Graphics Software

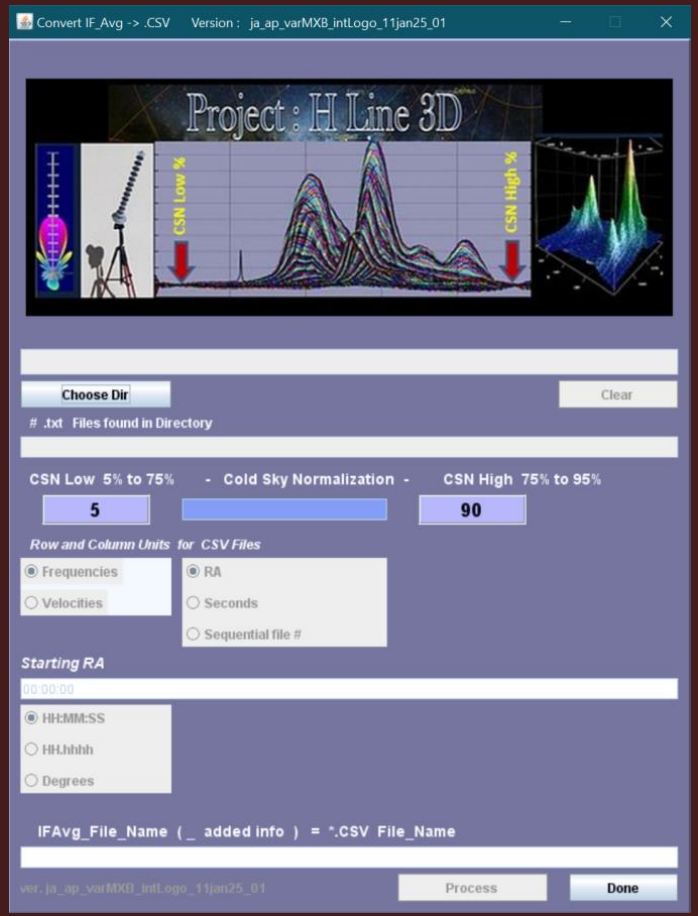
Rinearn 3D Graphics Software

From installation subdirectory , **create Shortcut** to *RinearnGraph3D_5.6.36.bat*
and Drag Shortcut to Desktop



Name	Date modified	Type	Size
AnimationInput	8/18/2024 6:13 AM	File folder	
AnimationOutput	8/18/2024 6:13 AM	File folder	
bin	8/18/2024 6:13 AM	File folder	
etc	8/18/2024 6:13 AM	File folder	
jre	8/18/2024 6:19 AM	File folder	
lib	8/18/2024 6:13 AM	File folder	
License	8/18/2024 6:13 AM	File folder	
RinearnGraph3DProgram	8/18/2024 6:13 AM	File folder	
RinearnGraph3DQuickSetting	8/18/2024 6:13 AM	File folder	
Sample	8/18/2024 6:13 AM	File folder	
ReadMe.txt	8/18/2024 6:13 AM	Text Document	8 KB
RinearnGraph3D.jar	8/18/2024 6:13 AM	JAR File	220 KB
RinearnGraph3D_5.6.36.bat	8/18/2024 6:13 AM	Windows Batch File	1 KB
RinearnGraph3D_5.6.36.bat - Shortcut	8/18/2024 6:20 AM	Shortcut	2 KB
SetMemorySize.bat	8/18/2024 6:13 AM	Windows Batch File	1 KB
UpdateJRE.bat	8/18/2024 6:13 AM	Windows Batch File	1 KB

JAVA_HL3D_ . jar Processing Software Installation



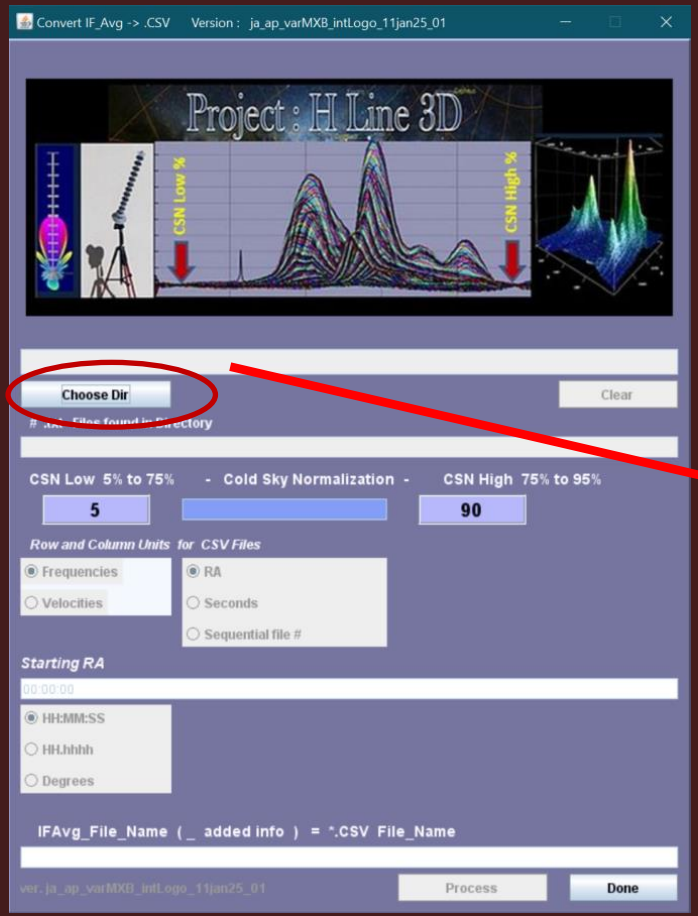
Conversion of a subdirectory data set
of IF_Ave *.txt files to Rinearn compatible *.csv

https://github.com/AP-HLine-3D/HLine3D/

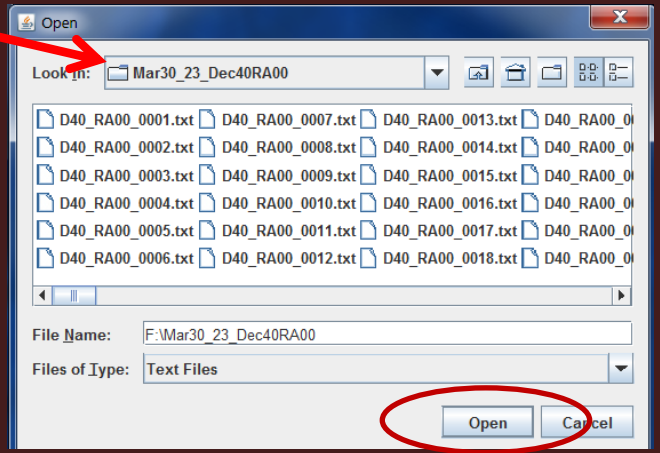
download Latest Rev of JAVA_HL3D_.jar and drag to Desktop
(see ***slide #3*** regarding language compatibility installation)

JAVA_HL3D_33%Me-Mi_Vmxb_Vsplt_14May26.jar

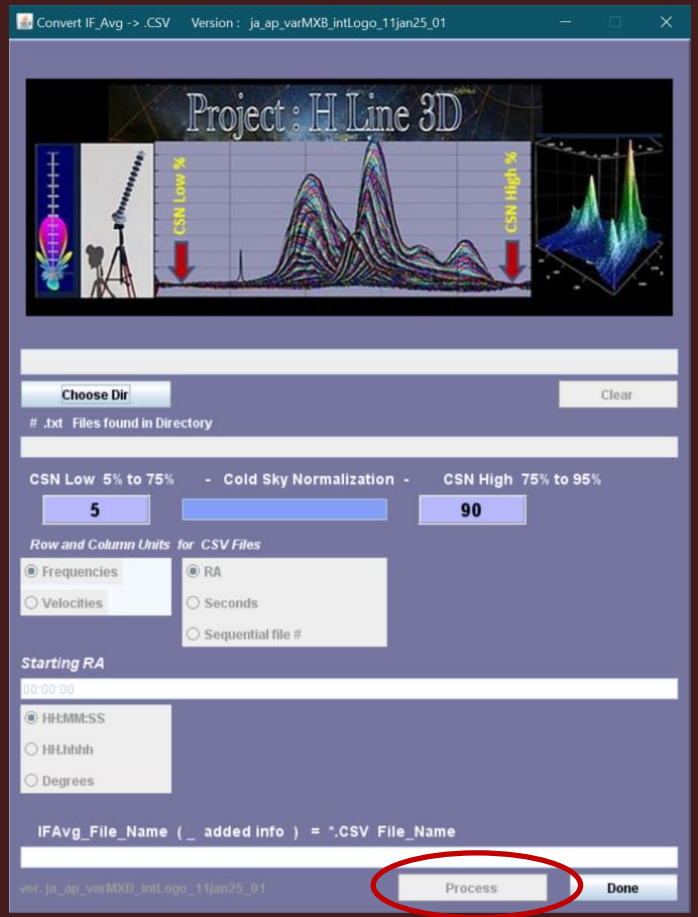
JAVA_HL3D_ . jar Processing Software Guide



1) Select ***'Choose Dir'*** *,* ***Fileset and*** ***'Open'***

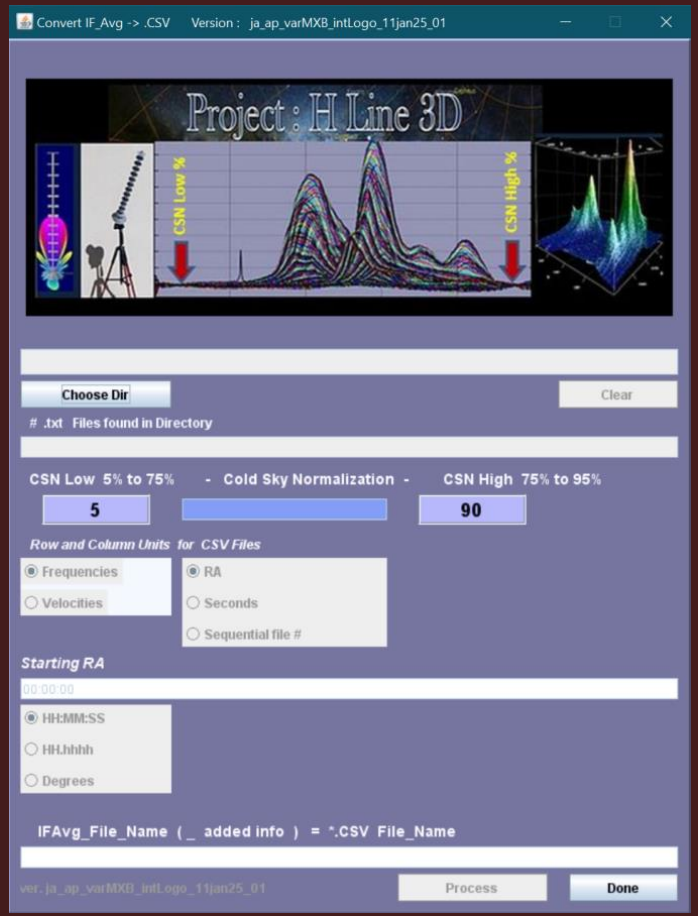


JAVA_HL3D_ . jar Processing Software Guide

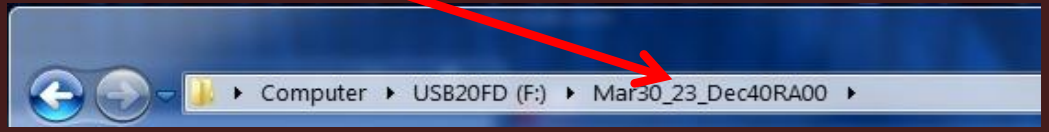


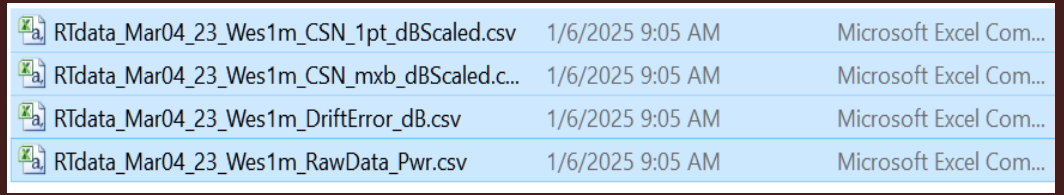
2) (optional: append info to) 'IFAvg_File_Name' and 'Process'

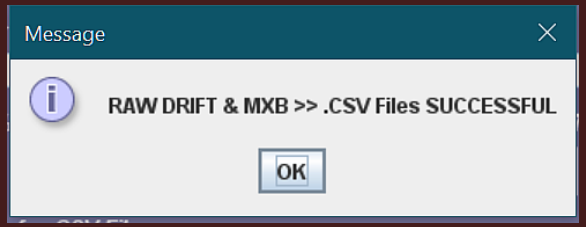
JAVA_HL3D_ . jar Processing Software Guide



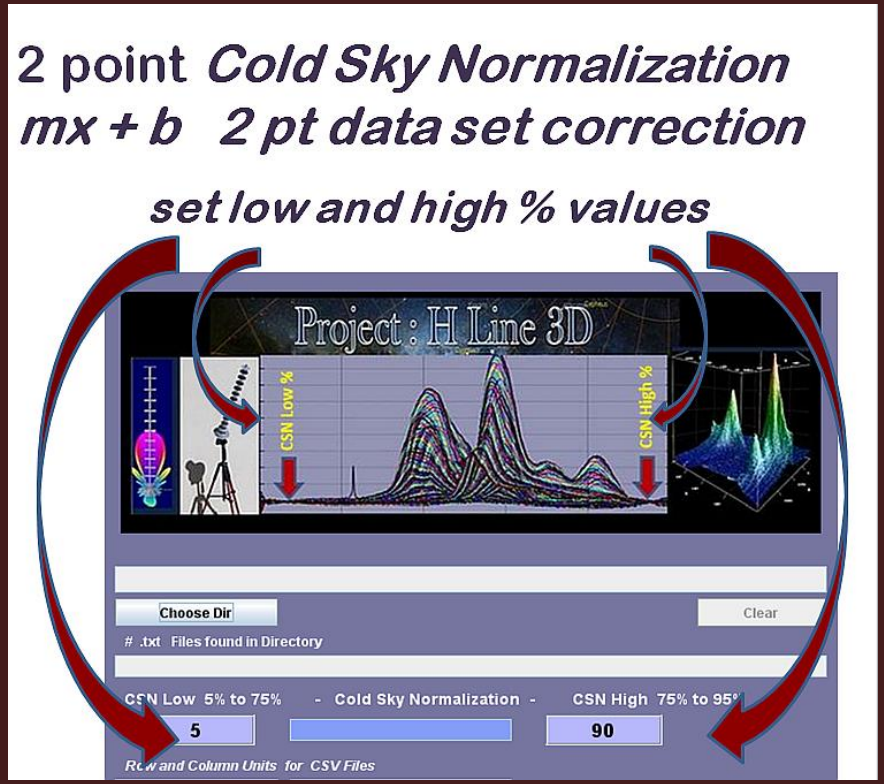
3) *.CSV files will be written to original Fileset Directory







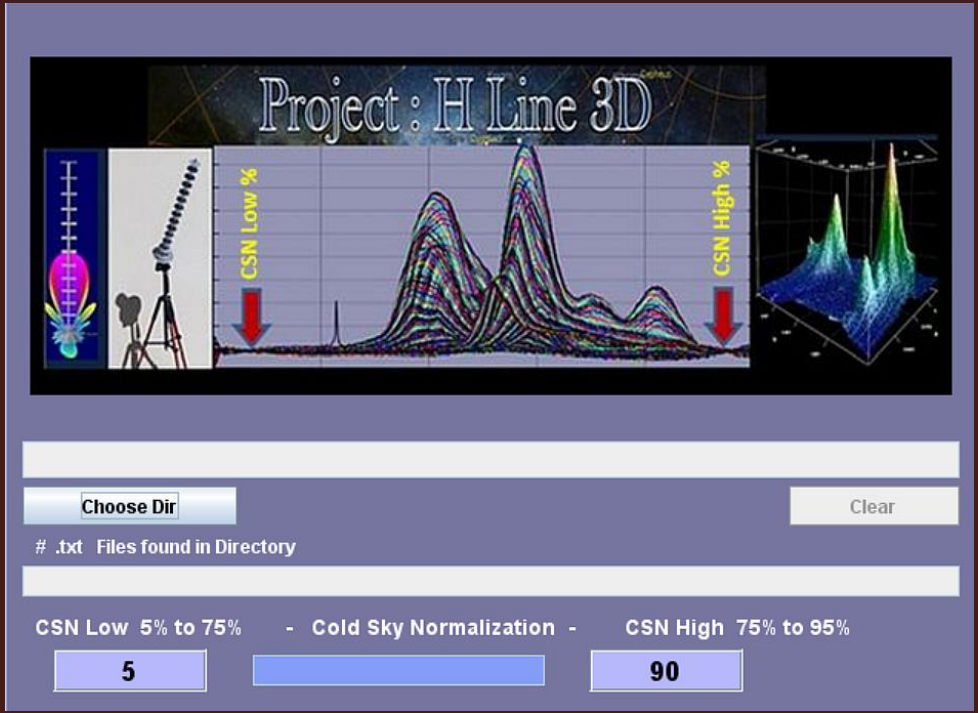
JAVA_HL3D_ . jar Processing Software Guide

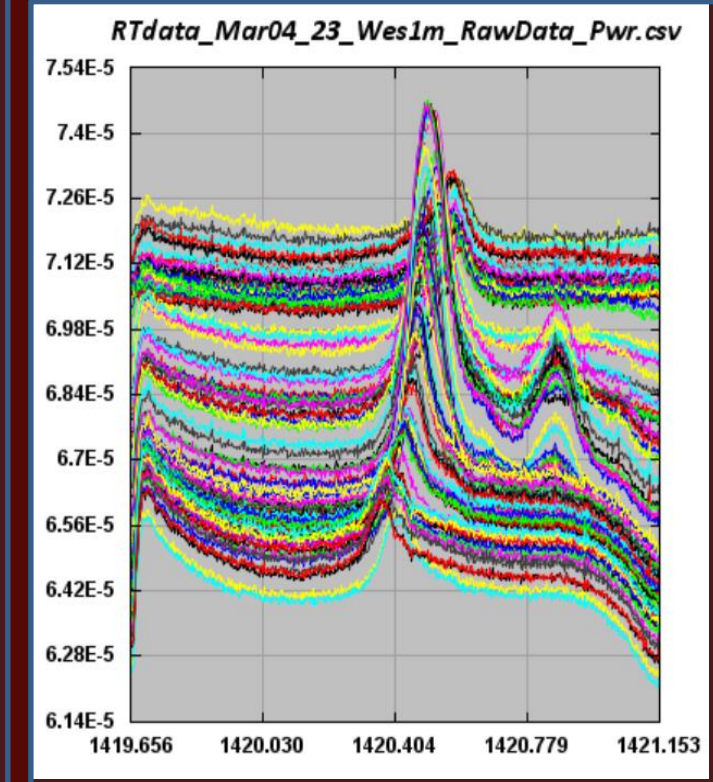


Enter %Low and %High Frequencies
for Cold_Sky_Normalization (CSN)

JAVA_HL3D_ . jar Processing Software Guide

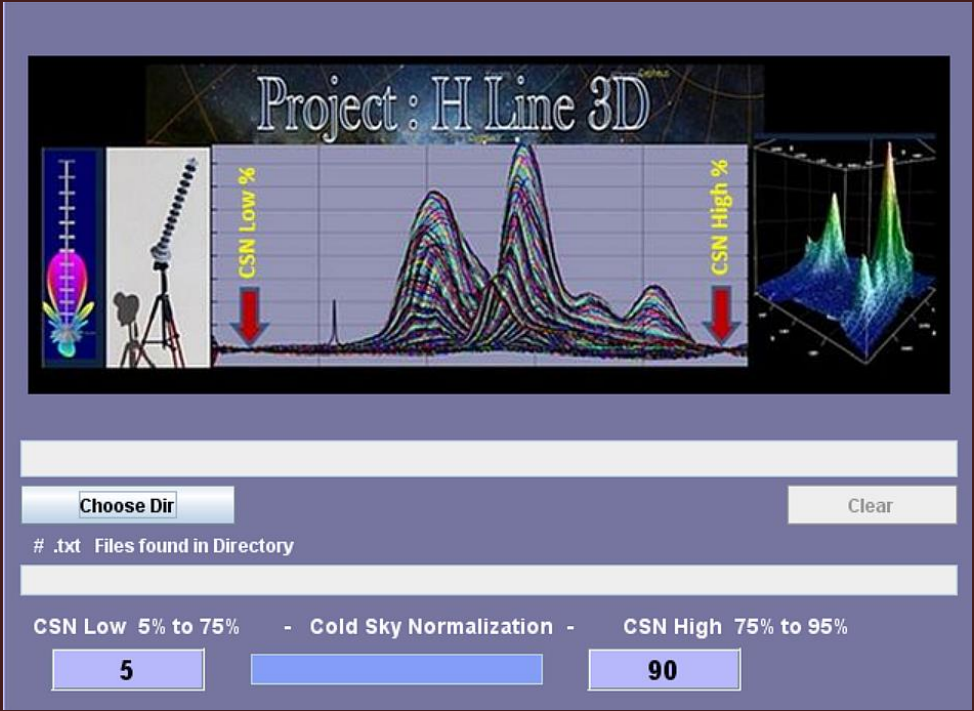
1) Raw Uncorrected Data containing frame to frame drift

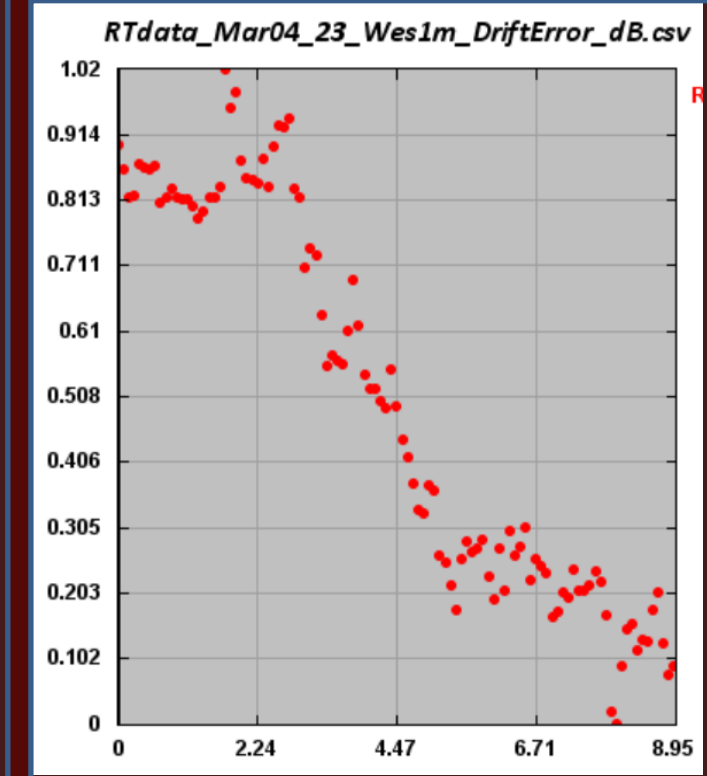




JAVA_HL3D_ . jar Processing Software Guide

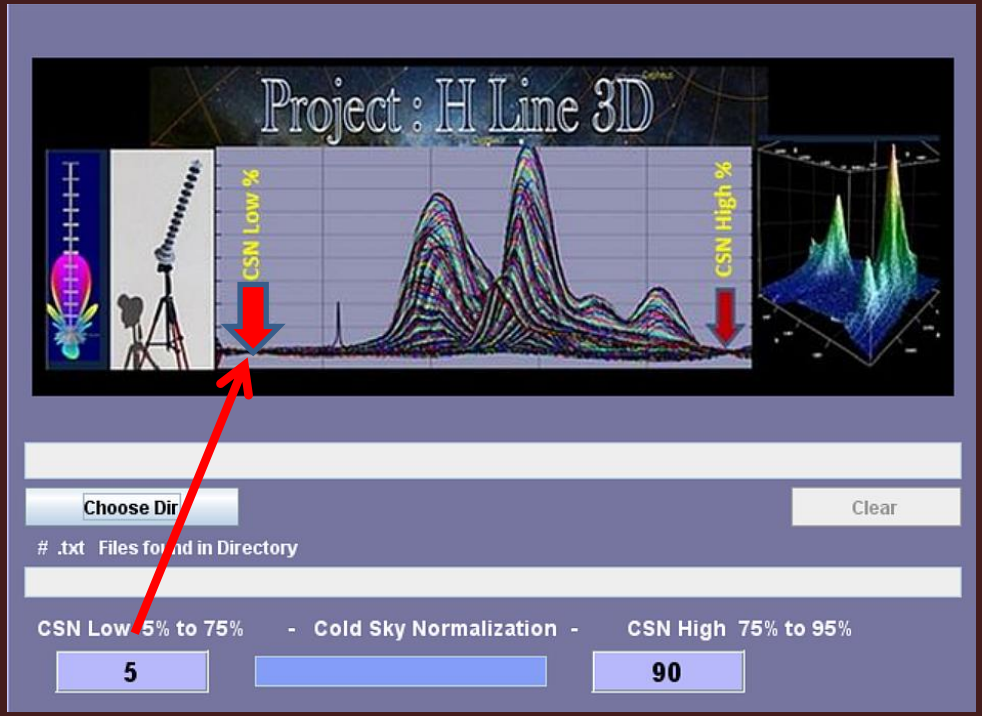
2) Frame to Frame drift
X axis Hours of Time

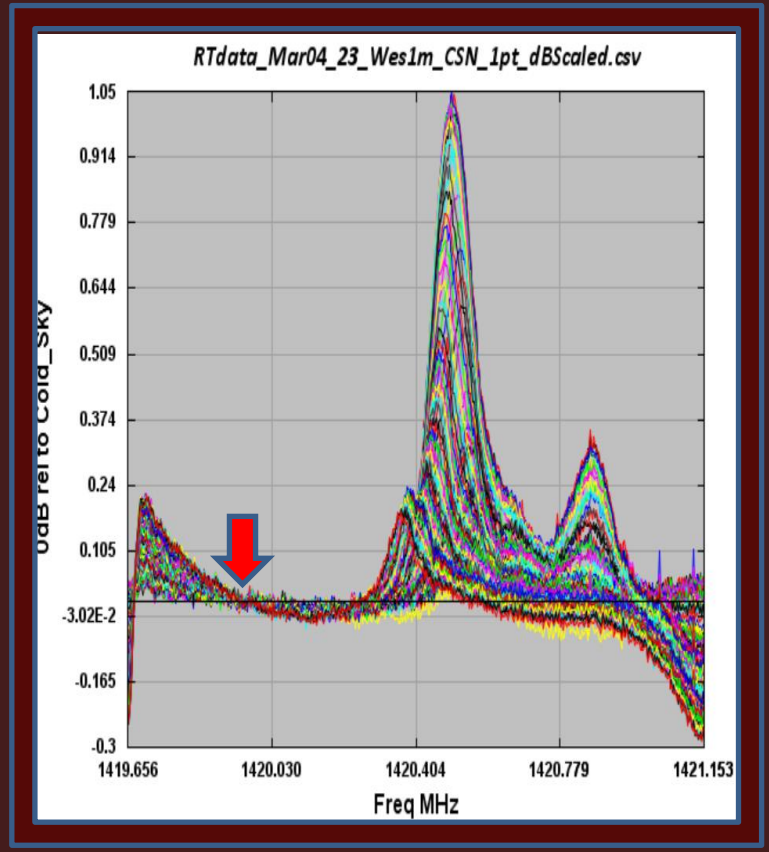




JAVA_HL3D_ . jar Processing Software Guide

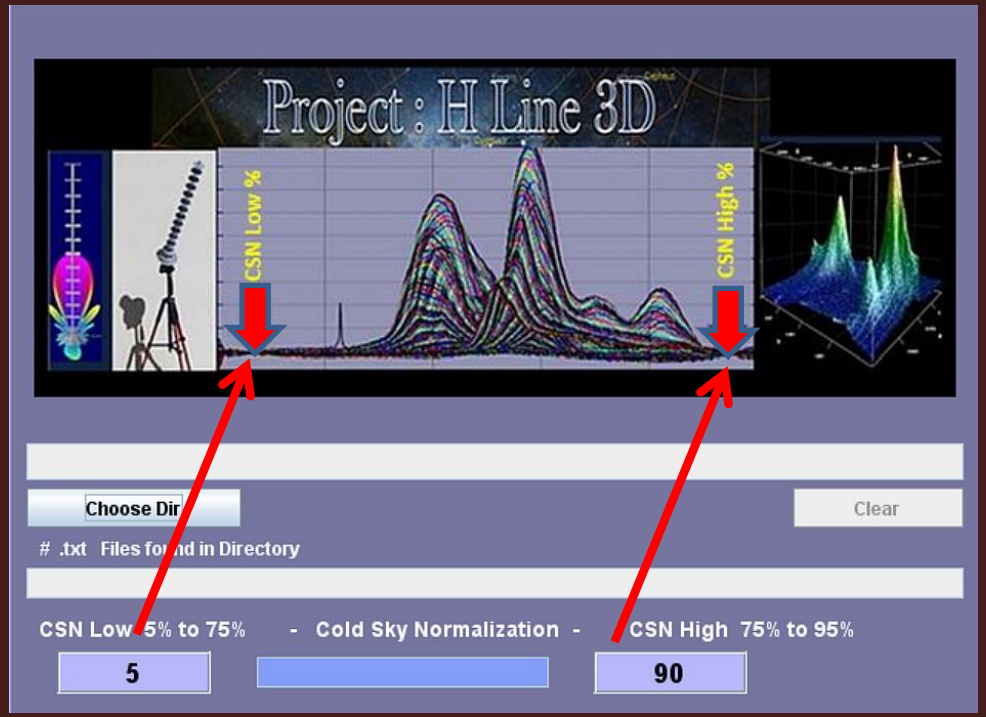
3) %Low Single 1pt Linear Drift
CSN normalized

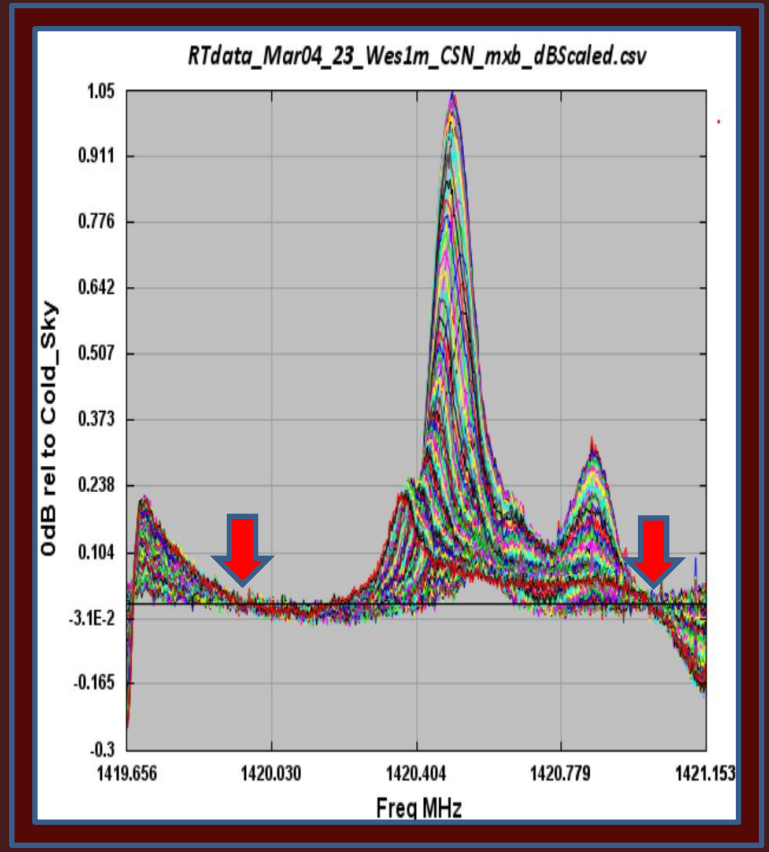




JAVA_HL3D_ . jar Processing Software Guide

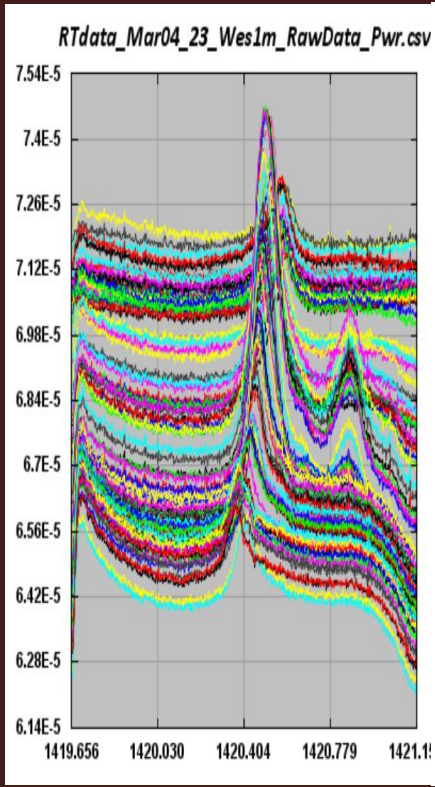
4) %Low and %High 2pt mx+b Slope
CSN normalized



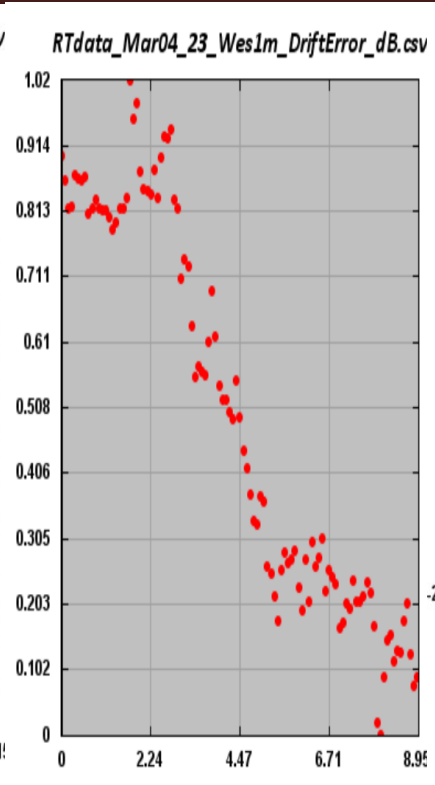


JAVA_HL3D_ . jar Processing Software Guide

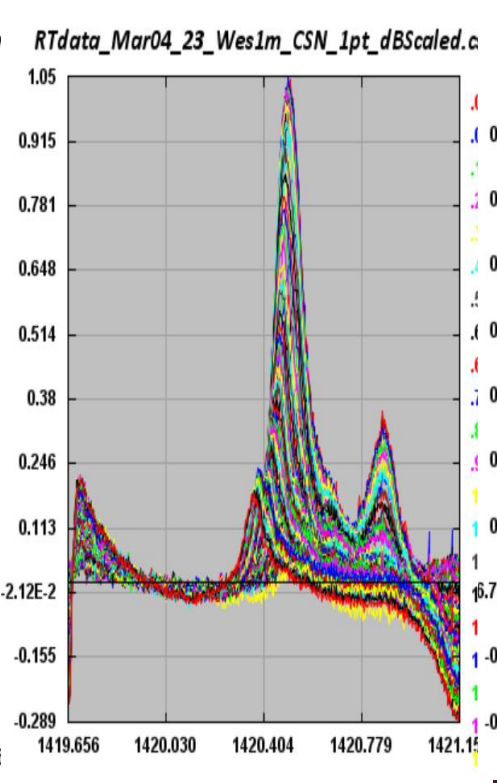
1) **Raw Data containing frame to frame drift**



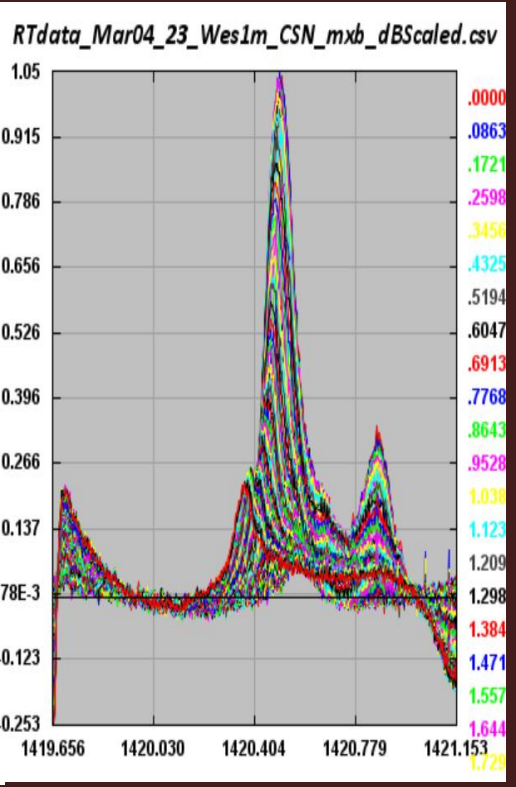
2) **Frame to Frame Drift X axis Hours of Time**



3) **%Low 1pt Linear CSN normalized**

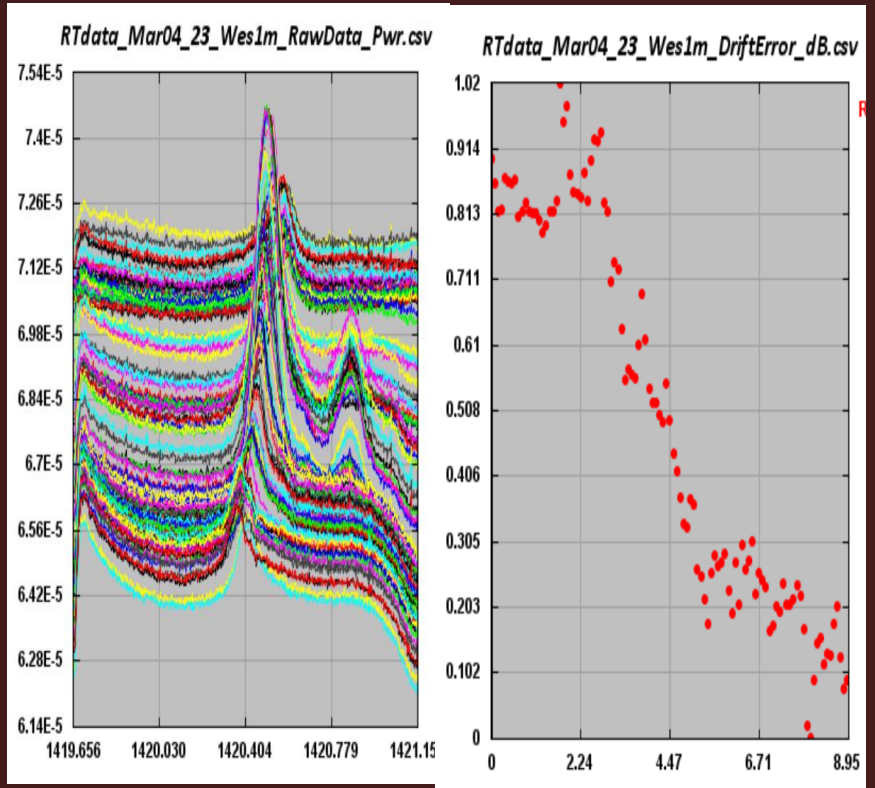


4) **%Low & %High 2pt mx+b Slope CSN normalized**



JAVA_HL3D_ . jar Processing Software Guide

" Cold Sky Normalization" 1 pt Linear Background Drift Correction



During the time period in which spectral data is being acquired, many factors may cause an overall background level shift …

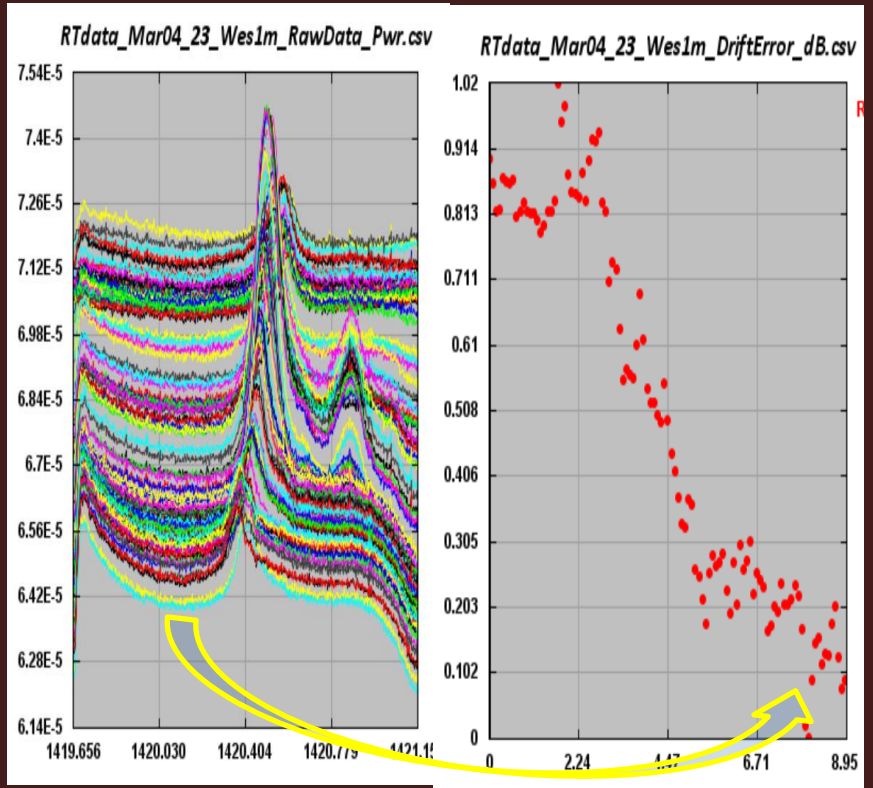
1) Atmospheric Transparency
2) Sunlight
   being received in the antenna's beam side-lobe patterns
   or being reflected off nearby vegetation
   (trees, buildings, bushes)
3) Temperature Changes
   in the surrounding environment
   or heating of the antenna and electrical components

FYI : The nooelec SAWBird H1 LNA has a thermal amp gain drift of only **-0.1 dB / +10 dgC .**

As such, the corrections are considered changes in noise level and the drift is *subtracted from* the data referenced to the lowest value rather than *divided by* as would be done for correcting amplifier gain change.

JAVA_HL3D_ . jar Processing Software Guide

" Cold Sky Normalization" 1 pt Linear Background Drift Correction



The Background Drift can be measured and plotted.

Step #1 Specify a frequency (%Low) below than any Hydrogen Line data with which to 'Normalize' the Drift This can be considered the " Cold Sky " Background

Step #2 Find the spectrum with the Lowest Amplitude at that range and use it as the Drift Correction Reference.

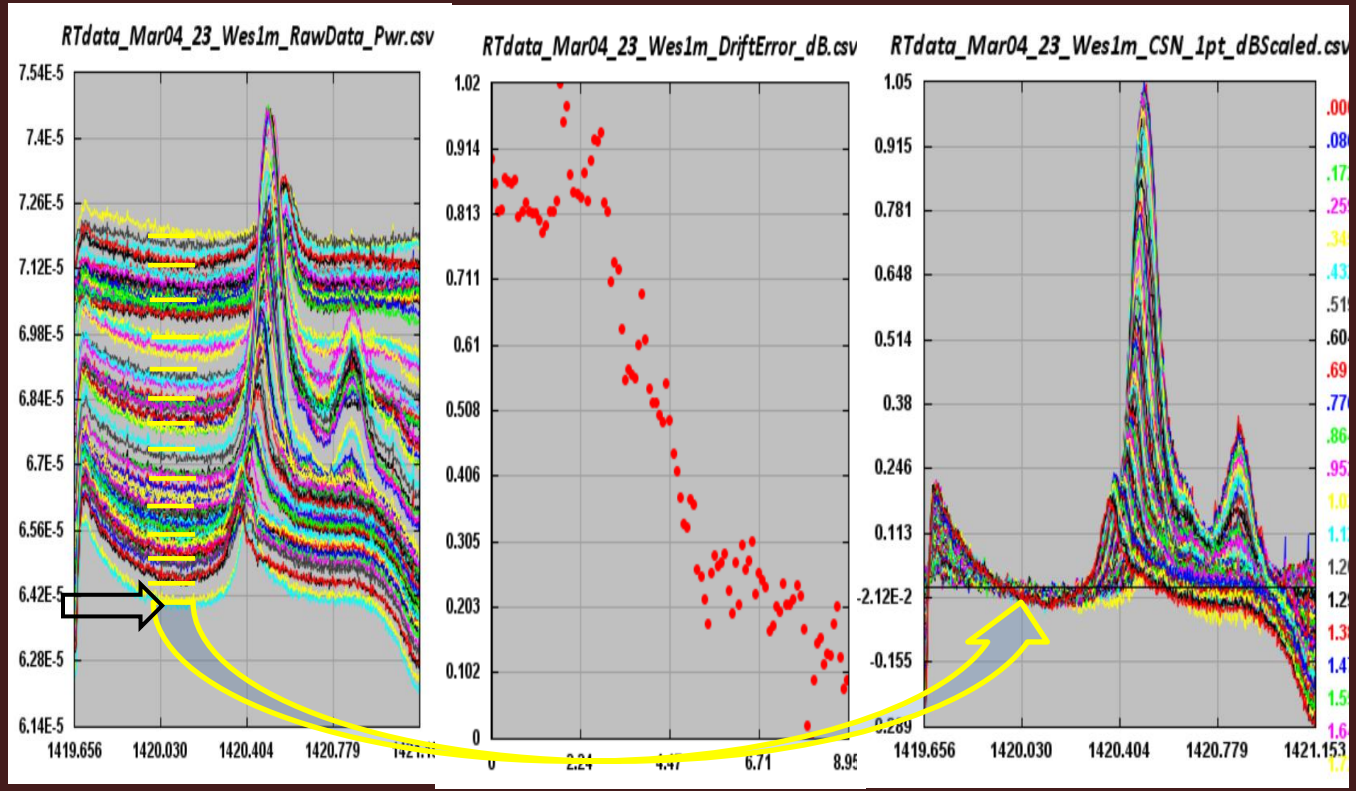
Step#3 For each spectra, calculate the difference in amplitude between its average value over the drift correction frequency range and the Reference, and subtract that from all values in that spectrum

Step#4 Repeat that for each of the spectra.

Lowest Amplitude = Linear Drift Correction Reference

JAVA_HL3D_ . jar Processing Software Guide

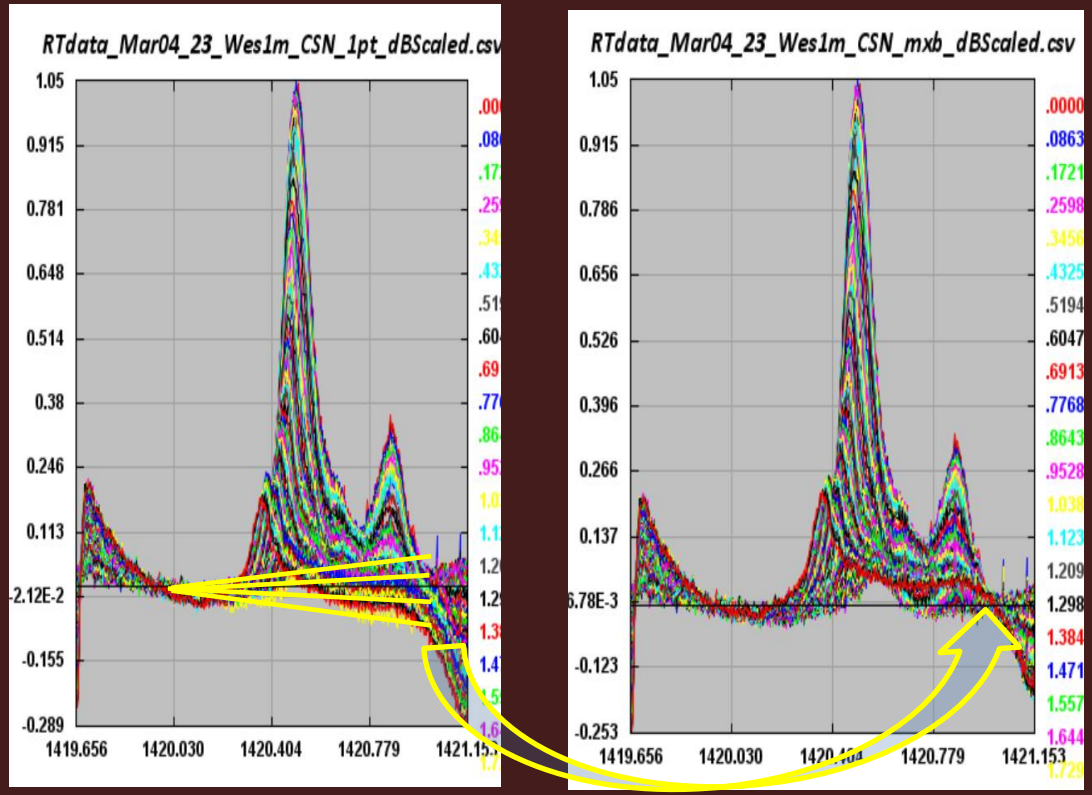
" Cold Sky Normalization" 1 pt Linear Background Drift Correction



" Cold-Sky" Drift Corrected Spectrum Set

JAVA_HL3D_ . jar Processing Software Guide

" Cold Sky Normalization" 2 pt mx+b Slope Background Drift Correction



Step #1 Specify the %High frequency above any Hydrogen Line data with which to 'Normalize' And correct the Slope Drift
This is also the " Cold Sky " Background

Step #2 Find the difference of each spectrum above or below the " Cold Sky "

Step#3 For each spectra, calculate an mx+b Slope correction and subtract that from the spectrum error

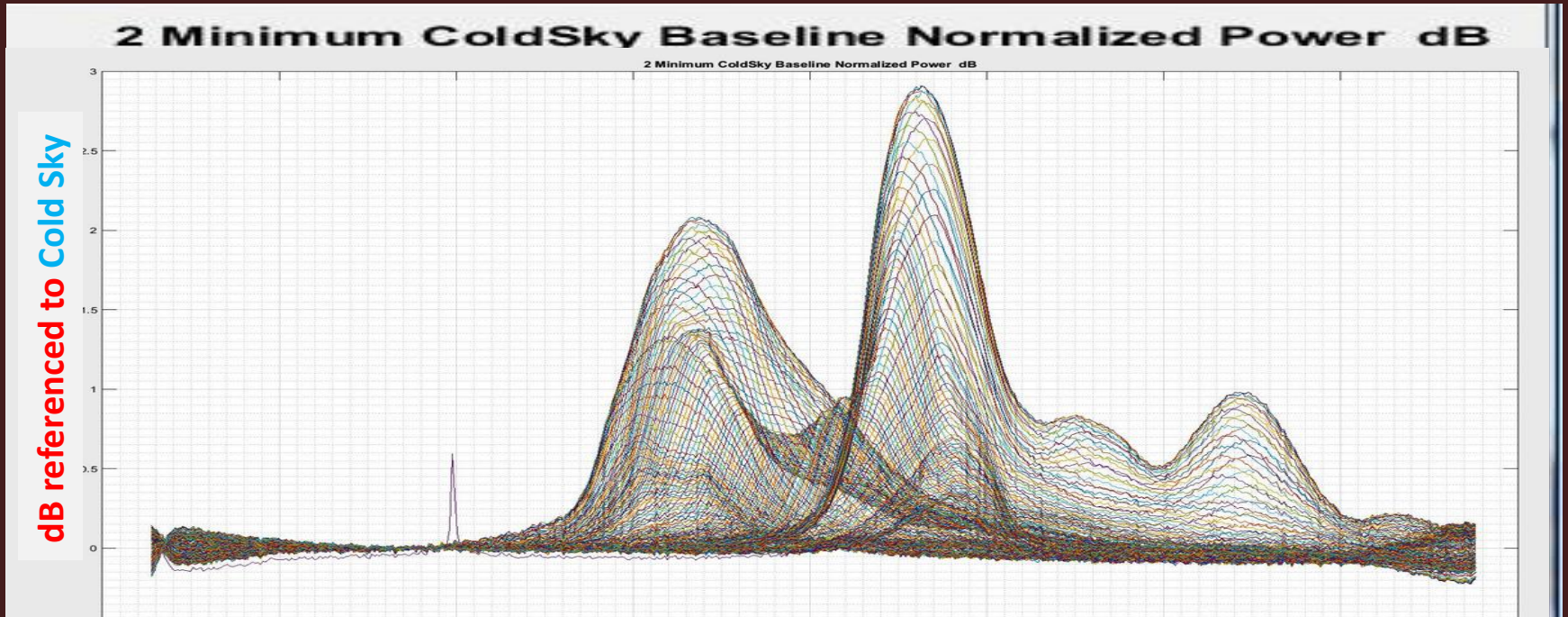
Step#4 Repeat that for each of the spectra.

“ Cold Sky ” Background Drift Corrected Spectrum Set Scaled in dB

A Decibel Scale is a Logarithmic Scale which enables data with amplitudes
over many orders of magnitude to be displayed with ‘nearly equal’ *physical size* on a plot

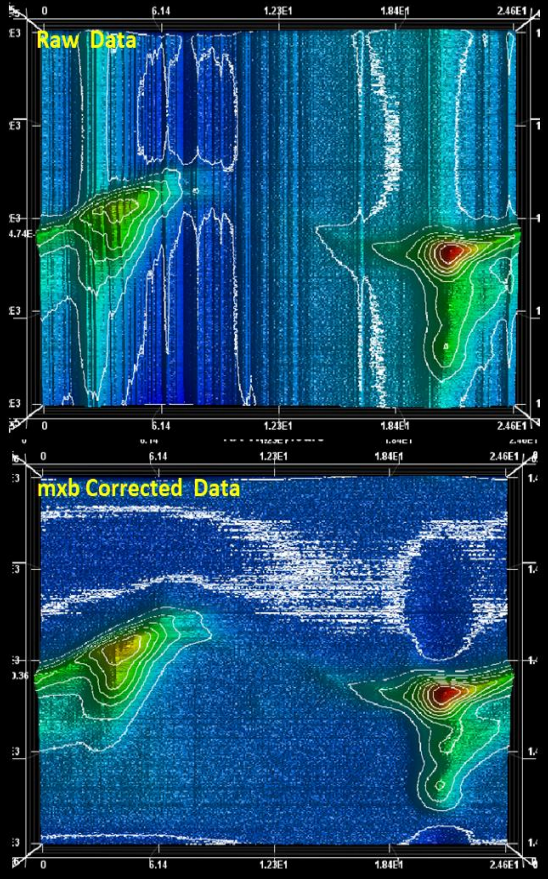
The conversion equation from Power to dB (Decibel) is **dB = 10 * Log10 (power value)**
As a reference, the normalized Cold_Sky Power Amplitude is set to equal Zero dB

The Vertical Y axis on this graph is labeled : **dB referenced to Cold Sky**

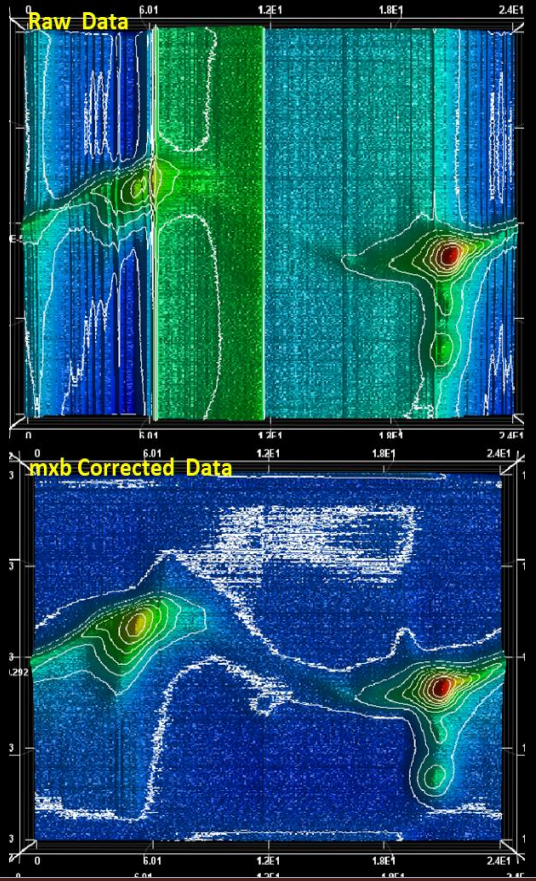


Examples : *Raw* vs *2 pt mx+b Slope Cold_Sky_Normalization Correction*

PRT 1.2m Dish *" ~ normal data "*



PRT 1.2m Dish *added plastic sheet dew shield @ 5-10 hrs*



0.75m Disk Yagi *high Sun noise bleedover into side lobe*

